



Mahmoud Ayman Waheed

AI Engineer

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atosfit.com

SUMMARY

Results-driven AI Engineer and recent Bioinformatics & Artificial Intelligence graduate from Delta University for Science and Technology (CGPA 3.0/4.0). Awarded 2nd Prize (\$3,000) at Huawei Developer Competition 2025 (Northern Africa) and ranked Top 30 Globally among 12,000+ projects in the World Computer Hacker League for developing AtosFit, a real-time computer vision fitness coaching platform. Proven expertise in Computer Vision, deep learning architectures (YOLOv9, kinematic pose estimation), edge AI, embedded robotics (ESP32, Arduino), and end-to-end full-stack AI deployment. Passionate about solving complex real-world challenges through intelligent, scalable autonomous systems.

WORK EXPERIENCE

Lead AI Engineer & Founder, AtosFit (atosfit.com) Oct 2024 – Present

- **2nd Prize Winner (\$3,000 Award) at Huawei Developer Competition 2025 (Northern Africa Finals) & Top 30 Globally** among 12,000+ projects in **World Computer Hacker League (WCHL 2025)**.
- Architected and deployed sub-30ms joint angle kinematic pose estimation for real-time exercise form correction and automated repetition counting.
- Engineered 100% on-device edge AI processing, ensuring complete user video privacy without transmitting video feeds to external cloud servers.
- Integrated a 24/7 conversational fitness coach and automated nutrition generator; launched responsive web platform serving active users at atosfit.com.

AI & Robotics Developer, Quadpod Security Robot Feb 2024 – Jun 2024

- Engineered an autonomous 4-legged quadrupedal robot equipped with an ESP32-CAM optical sensor and edge-deployed YOLOv9 deep learning API for real-time weapon and concealed threat detection.
- Calibrated multi-axis servo gait kinematics and automated alert telemetry designed for high-security transit hubs and educational campuses.
- Achieved 96.8% detection accuracy with sub-40ms edge inference under dynamic indoor lighting and partial occlusions.

Embedded Systems Developer, Autonomous Firefighter Robot Oct 2023 – Jan 2024

- Developed an autonomous search-and-extinguish response vehicle utilizing Arduino Uno, dual IR flame sensors, and HC-SR04 ultrasonic collision avoidance.
- Integrated an isolated relay-driven high-pressure DC water pump and built a custom Bluetooth mobile telemetry app for remote steering and override.

EDUCATION

Bachelor of Science in Artificial Intelligence Oct 2022 – Jul 2026

Delta University for Science and Technology — Faculty of AI

- Major: Bioinformatics **Final Cumulative GPA: 3.0 / 4.0**
- Graduation Project: AtosFit — AI Vision Coach **(Grade: Excellent · First Class Honors)**

KEY SKILLS

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|-------------------------------------|-----------------------------------|--|
| • Computer Vision & OpenCV | • Python & C++ Programming | • Embedded Systems (ESP32/Arduino) |
| • YOLOv9 Object Detection | • React.js & TypeScript FastAPI & | • Robotics Kinematics & GAIT Sensor |
| • Kinematic Pose Estimation | • RESTful APIs | • Fusion & Telemetry Problem-Solving & |
| • Deep Learning & PyTorch Edge AI & | • Git, GitHub & CI/CD | • Innovation Technical Stage |
| • Model Optimization | • AI Agent & LLM Integration | • Presentation |